Design of A Bulk-Micromachined Piezoelectric Accelerometer

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***Abstract*—A bulk-micromachined piezoelectric accelerometer is proposed and designed, in which the bulk-micromachined silicon seismic mass is suspended by the cantilevers and part of cantilever is covered with ZnO piezoelectric thin film. In this paper, we first derive the one-dimensional analytical model of the piezoelectric cantilever to study the factors affecting the sensitivity of the accelerometer, and then design the piezoelectric accelerometers with straight and folded cantilever structures. Totally, 12 different structures of the microaccelemeter are investigated. The results show that this type of bulk-micromachined piezoelectric accelerometer can own high sensitivity and wide working range to meet requirement of some applications in aerospace and mechanical shock inspection for vehicles.**

Keywords- MEMS; bulk-micromachining; piezoelectric microaccelerometer; zinc oxide

I. INTRODUCTION

The microaccelerometer are widely used in many applications, such as automobiles, consumer electronics, biomedical instrumentations, due to its small size, low power consumption and low cost. However, most of current microaccelerometers use polysilicon thin film layer and are fabricated into the sensing element by surface micromachining technology, leading to limited sensitivity and low ability to bear huge mechanical shock. Therefore, for some applications, such as space navigation and monitoring mechanical shock and vibration during transportation, microaccelerometers with high sensitivity and high ability to bear mechanical shock are needed. In this paper, we present and deign a bulk-micromachined piezoelectric accelerometer with different structures for these special applications.

II. FUNDAMENTALS OF PIEZOELECTRIC ACCELEROMETER

The schematic diagram of piezoelectric accelerometer is shown in Fig.1. When the cantilever is forced with a vertical load in the center of mass block, the deflection equation of the cantilever can be expressed as

$$\varepsilon(x) = \frac{3(l_c + l_m/2)x^2 - x^3}{6EI_y} F_z \qquad 0 \le x \le l_c \quad (1)$$

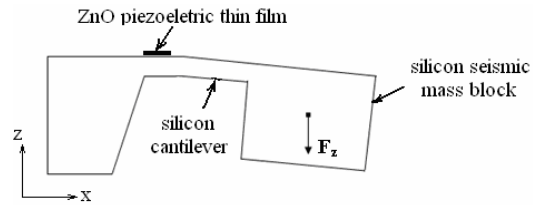


Figure 1. Working principle of the piezoelectric accelerometer

where E is Young's modulus, I_y the rotary inertia of the cantilever,, l_c and l_m the length of cantilever and mass block, respectively. And, the deflection δ and angle θ at $x=l_c$ are

$$\delta = \varepsilon(l_b) = \frac{4l_c^{\,3} + 3l_c^{\,2}l_m}{12EI_y} F_z \quad (2)$$

$$\theta = \varepsilon'(l_b) = \frac{l_c^{\,2} + l_c l_m}{2EI_y} F_z \quad (3)$$

When the accelerometer works, the force acting on mass block leads to deflection of cantilever, and maximum stress occurs at the fixed end of the cantilever. It is therefore the ZnO piezoelectric thin film is sputtered and patterned laid on the fixed end of the cantilever, generating an electric potential that is proportional to the force applied on the mass block, i.e. the acceleration.

The influences of piezoelectric cantilever structure design for accelerometer sensitivity are studied at first. When the cantilever with a piezoelectric thin film on it deflects, the output voltage from the piezoelectric thin film layer can be derived as

$$e_a = \frac{-3d_{31}S_{11}^{Si}S_{11}^{P}h_{Si}h_P(h_{Si}+h_P)l_c}{\left[K\varepsilon_{33}^{T} - d_{31}^2 h_{Si}(S_{11}^{Si}h_P^{\,3} + S_{11}^{P}h_{Si}^{\,3})\right]b_c} F_z \quad (4)$$

and the sensitivity of the piezoelectric accelerometer can be expressed as

$$S_u = \frac{e_a}{a} = \frac{-3d_{31}S_{11}^{Si}S_{11}^{P}h_{Si}h_P(h_{Si}+h_P)l_c}{\left[K\varepsilon_{33}^{T} - d_{31}^2 h_{Si}(S_{11}^{Si}h_P^{\,3} + S_{11}^{P}h_{Si}^{\,3})\right]b_c} M \quad (5)$$

where the constant K is expressed as

$$K = (S_{11}^{Si})^2 (h_P)^4 + 4S_{11}^{Si} S_{11}^{P} h_{Si} (h_P)^3 + 6S_{11}^{Si} S_{11}^{P} (h_{Si})^2 (h_P)^2 + 4S_{11}^{Si} S_{11}^{P} h_P (h_{Si})^3 + (S_{11}^{P})^2 (h_{Si})^4 \quad (6)$$

where S_{11}^{Si} and S_{11}^{P} are the elastic compliance constant of silicon and ZnO, respectively. h_{Si}, h_P, l_c and b_c thickness of silicon and ZnO, length and width of the cantilever, respectively, d_{31} the piezoelectric constant, and ε_{33}^{T} the dielectric coefficient.

The maximum stress along x direction distributes at the fixed end is

$$\sigma = M_r / W \quad (7)$$

where M_r and W are the bending moment and bending resistance section modulus of the cantilever, respectively, and

$$M_r = Mal_c \quad (8)$$

$$W = b_c h_{Si}^{\ 2} / 6 \quad (9)$$

The strain at the fixed end of silicon cantilever can be obtained by

$$\varepsilon = \sigma / E \quad (10)$$

Generally, the maximum strain of silicon material is 10^3, but in practice, the strain endured by the fixed end of silicon cantilever is required to less than 5×10^{-4} in order to ensure the good linearity of output.

In our design, the size of the mass block is $1000\mu m\times1000\mu m\times400\mu m$. According to Eq.(5), sensitivity is proportional to the ratio of length to width of the cantilever. In order to ensure that the accelerometer owns a large measurement scale, the ratio of length to width of the cantilever cannot be too large, and we set this ratio at two. Given the above parameters, the relationship between sensitivity and the thickness of ZnO piezoelectric thin film layer is shown in Fig.2, indicating that the sensitivity increases along with the increasing of the thickness of ZnO layer. However, the thickness of ZnO layer is limited by microfabrication technology. With the magnetron sputtering technology used for fabrication of our proposed accelerometer, the maximum value of thickness of ZnO layer is $2.2\mu m$.

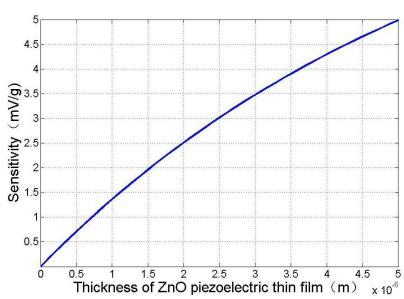


Figure 2. The relationship between the accelerometer sensitivity and the thickness of ZnO piezoelectric thin film layer

III. Analysis of The Bulk-Micromachined Piezoelectric Microaccelerometer

First, we study the microaccelerometer with straight line cantilever structures. 6 different structures of this form of the microaccelerometers are investigated, which are 1-cantilever, 2-cantilever, 4-straight cantilever for one and dual-seismic mass block. The cantilevers and mass block are made by bulk-etched silicon, and the ZnO thin film is sputtered and patterned at the fixed end of the cantilever. According to the analysis in above section, the structural parameters of the microaccelerometer are shown in Table I.

Finite element method in ANSYS is used to analyze microaccelerometer with different structures and the distribution of strain on the cantilever under different acting forces. With the requirement of the strain endured by the fixed end of silicon cantilever less than 5×10^{-4}, the measurement scales of microaccelerometer with different structures are obtained by FEM in ANSYS and shown in Table II. It indicates that with the bulk micromachined thick cantilevers, the accelerometers can bear a huge mechanical shock as high as 22000g of acceleration.

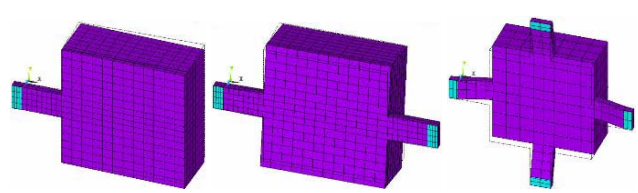
The fundamental mode shape and its resonance frequency of different structures of microaccelerometer can be obtained by performing modal analysis in ANSYS, shown in Fig. 3 and 4, respectively. Among these 6 structures, the working bandwidth of 4-straight cantilever for one seismic mass block is largest, which suggests that it can endure a mechanical shock at a frequency as high as ~112KHz.

TABLE I. Structure Parameters of The Micro-Accelerometer with Straight Cantilevers

Length of cantilever	Width of cantilever	Thickness of silicon	Length of mass block
400μm	200μm	50μm	1000μm
Thickness of mass block	Thickness of ZnO	Length of ZnO	Width of ZnO
400μm	2.2μm	80μm	200μm

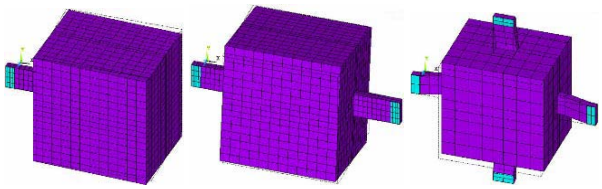
TABLE II. Measurement Scale of The Micro-Accelerometer with Straight Cantilevers

Structures	**Single mass block**			**Dual mass blocks**		
	1-canti-lever	***2-canti-lever***	***4-canti-lever***	***1-canti-lever***	***2-canti-lever***	***4-canti-lever***
Mea. scale	3100g	9700g	22000g	1560g	5300g	11700g



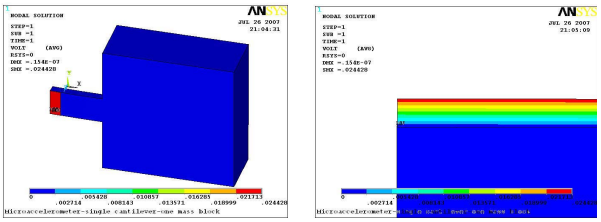
(a) f=8600Hz (b) f=35143Hz (c) f=111527Hz

Figure 3. The first order vibration shapes of straight cantilever structures with 1 mass block



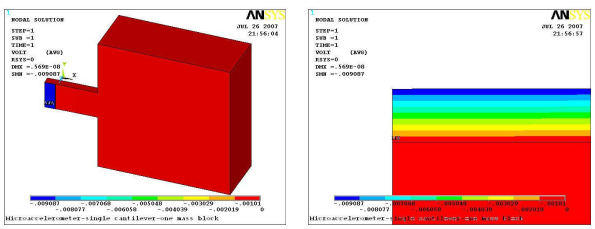
(a) f=6066Hz (b) f=24206Hz (c) f=81650Hz

Figure 4. The first order vibration shapes of straight cantilever structures with 2 mass blocks



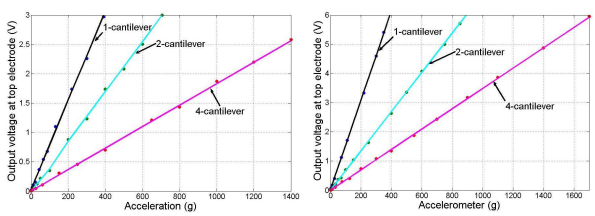
(a) (b)

Figure 5. The distribution of electric potential for 1-straight cantilever 1-mass block structure: (a) 3-dimensional view; (b) side view (The force is in the direction of gravity)



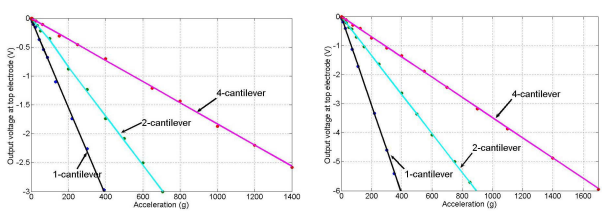
(a) (b)

Figure 6. The distribution of electric potential for 1-straight cantilever 1-mass block structure: (a) 3-dimensional view; (b) side view (The force is in the opposite direction of gravity)



(a) (b)

Figure 7. The relationship between output voltage and acceleration for straight cantilever structures (the force is in direction of gravity): (a) single mass block; (b) dual mass blocks



(a) (b)

Figure 8. The relationship of output voltage and acceleration for straight cantilever structures (the force is in opposite direction of gravity): (a) single mass block; (b) dual mass blocks

The distributions of electric potential are shown in Fig. 5 and Fig. 6, when the mass block is affected by a force of 20μN (the accelerometer is equal to 2.2g). We take the 1-cantilever for one mass block structure for example here.

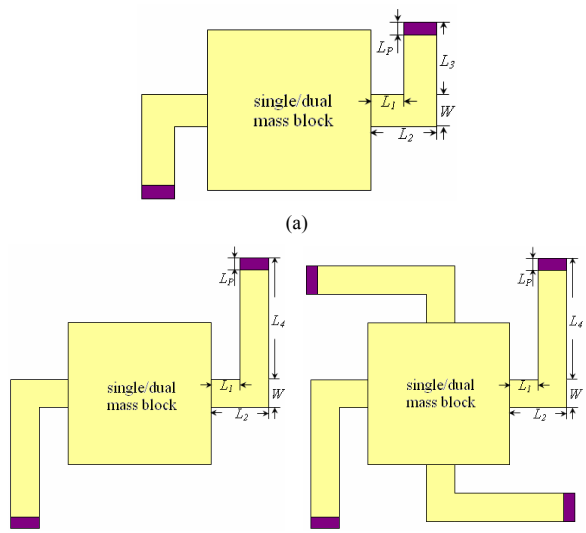
The relationship between output voltage and acceleration are obtained and shown in Fig. 7 and 8. The sensitivity of the microaccelerometer is the ratio of output voltage to acceleration that is the slope of lines obtained by calculation in Fig. 7 and Fig. 8, and results are listed in Table Ⅲ.

According to Table Ⅲ, the sensitivity of the accelerometer with 1-straight cantilever for one-seismic mass block structure is the largest, but its measurement scale is the lowest.

Furthermore, we design and analyze the bulk-micromachined accelerometers with the folded cantilever structures. The structural types and their parameters are shown in Fig. 9 and Table Ⅳ.

TABLE III. SENSITIVITY OF THE MICRO-ACCELEROMETER WITH STRAIGHT CANTILEVERS

Structures	Single mass block			Dual mass blocks		
	1-canti-lever	*2-canti-lever*	*4-canti-lever*	*1-canti-lever*	*2-canti-lever*	*4-cant-lever*
Sensitivity (mV/g)	7.67	4.24	1.83	15.32	6.72	3.48



(a)

(b) (c)

Figure 9. Schematic diagram of microaccelerometers with folded cantilever structures: (a) 2-cantilever (short); (b) 2-cantilever (long); (c) 4-cantilever.

TABLE IV. STRUCTURE PARAMETERS OF THE MICRO-ACCELEROMETER WITH FOLDED CANTILEVERS

Length of cantilever L_1	**Length of cantilever L_2**	**Length of cantilever L_3**	**Length of cantilever L_4**	**Length of ZnO L_p**
200μm	400μm	450μm	850μm	80μm
Thickness of silicon	**Length of mass block**	**Thickness of mass block**	**Width of cantilever**	**Thickness of ZnO**
50μm	1000μm	400μm	200μm	2.2μm

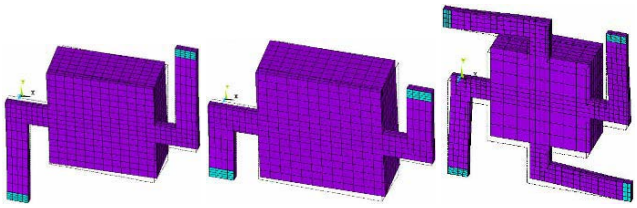
The measurement scales of the different folded cantilever structures are analyzed and the results are shown in Table Ⅴ. The fundamental mode shape and its resonance frequency of different folded structures of microaccelerometer are shown in Fig. 10 and 11, and the 4-cantilever structure for one seismic mass block still has the largest working bandwidth.

Similarly, the distributions of electric potential are obtained and shown in Fig. 12, and the relationship between output voltage and acceleration is shown in Fig. 13 and Fig. 14. The sensitivity of the microaccelerometer is listed in Table Ⅵ.

According to the results in Table Ⅵ, the microaccelerometer with 2-folded cantilever (short) for dual-mass block structure has the highest sensitivity. Compared to straight cantilever structure, the folded cantilever structure improves the sensitivity of the microaccelerometer, but the measurement scale decreases.

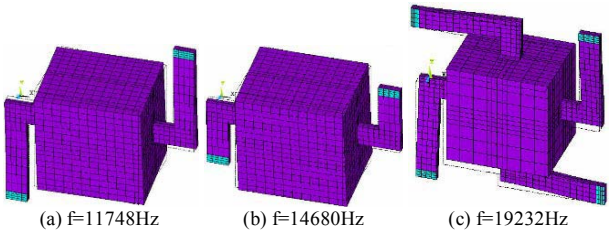
TABLE V. MEASUREMENT SCALE OF THE MICRO-ACCELEROMETER WITH FOLDED CANTILEVERS

Structures	Single mass block			Dual mass blocks		
	2-cantilever (long)	*2-cantilever (short)*	*4-cantilever*	*2-cantilever (long)*	*2-cantilever (short)*	*4-cantilever*
Mea. scale	4200g	6400g	8450g	2230g	3350g	4350g



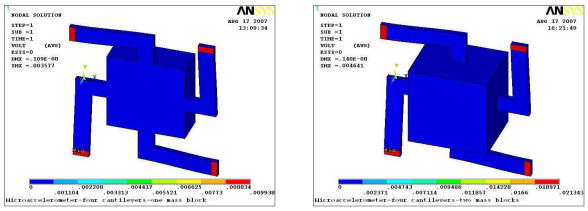
(a) f=17082Hz (b) f=21470Hz (c) f=26573Hz

Figure 10. The first order vibration shapes of folded cantilever structures with 1 mass block



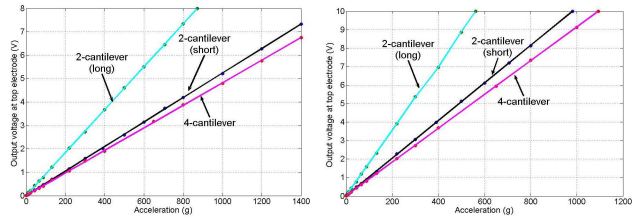
(a) f=11748Hz (b) f=14680Hz (c) f=19232Hz

Figure 11. The first order vibration shapes of folded cantilever structures with 2 mass block



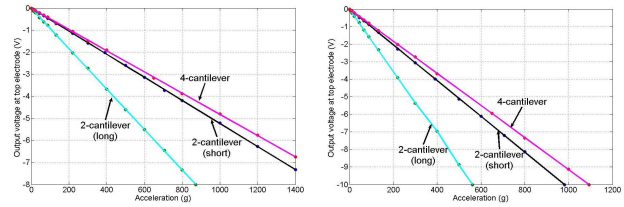
(a) (b)

Figure 12. The distribution of electric potential for 4-folded cantilever structure: (a) 1-mass block; (b) 2-mass block (The force is in the direction of gravity)



(a) (b)

Figure 13. The relationship of output voltage and acceleration for folded cantilever structures (the force is in direction of gravity): (a) single mass block; (b) dual mass blocks



(a) (b)

Figure 14. The relationship between output voltage and acceleration for straight cantilever structures (the force is in opposite direction of gravity): (a) single mass block; (b) dual mass blocks

TABLE VI. SENSITIVITY OF THE MICRO-ACCELEROMETER WITH FOLDED CANTILEVERS

Structures	Single mass block			Dual mass blocks		
	2-cantilever (long)	*2-cantilever (short)*	*4-cantilever*	*2-cantilever (long)*	*2-cantilever (short)*	*4-cantilever*
Sensitivity (mV/g)	9.16	5.23	4.82	17.76	10.18	9.14

IV. CONCLUSIONS

In this paper, we propose and design a bulk micromachined piezoelectric accelerometer. Totally, 6 straight cantilever structures and 6 folded cantilever structures of the microaccelemeters are investigated. The results show that this type of microaccelerometer can own high sensitivity and wide working range to meet requirements of some applications in aerospace and mechanical shock inspection for vehicles, and there is a trade-off between sensitivity and working range for this bulk micromachined piezoelectric accelerometer. For example, 1-straight cantilever structure microaccelerometer has the highest sensitivity but the lowest working range. On the contrary, 4-straight cantilever scheme has the lowest sensitivity but the highest working range. Different schemes can be used to different requirements.

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